

Financial News Sentiment and Stock Market Direction Prediction

Ajai Shakya Reem Ahmed Srivarshini Ankaiah Krishnappa

Primary model: FinBERT (ProsusAI) | Baseline: VADER

Text Mining LIS 501 | 2026

1,989

Trading Days

25

Headlines/Day

2008–2016

Date Range

53.5%

Up-Day Rate

Target Task and Problem

Efficient Market Hypothesis

*"Prices instantly reflect
all available information"*

VS

Behavioral Finance

*"Investors respond to
media tone and narrative"*

Thesis: Do daily Reddit news headlines contain extractable sentiment signals that predict DJIA price direction?

Five Research Questions



RQ1: Does aggregated daily news sentiment correlate with same-day DJIA direction?



RQ2: Can lagged sentiment predict next-day market movement above chance?



RQ3: Are negative headlines more influential than positive ones in driving market reactions?



RQ4: Do extreme sentiment days precede heightened volatility or larger price swings?



RQ5: Does news source domain matter — does general news produce weaker signals than financial news?

Related Work

Foundational 2007 — Tetlock

High negative word fraction in WSJ column predicts downward DJIA pressure

Lexicons 2011 — Loughran & McDonald

General word lists misclassify financial domain terms — domain-specific lexicons needed

VADER 2014 — Hutto & Gilbert

Rule-based tool for social media text — used here as simpler lexicon baseline

Comparison 2020 — Mishev et al.

Transformer gains over lexicons vary by corpus — motivates our source-domain RQ5

Firm-level 2008 — Tetlock et al.

Negative language in firm news predicts lower earnings realizations

Social Media 2011 — Bollen et al.

Twitter mood states Granger-cause DJIA movements, effects propagate over days

FinBERT ★ 2019 — Araci

Transformer fine-tuned on financial news and earnings calls — our primary model

Methods

1,989

Trading Days

25

Headlines/Day

2008–2016

Date Range

53.5%

Up Days

Analysis Pipeline

01

Preprocess

Strip b" artifacts, remove URLs, decode HTML entities

03

VADER Scoring

Baseline: score same text with rule-based VADER — neg, neu, pos, compound + per-headline avg

05

Time Split

80% train (2008–2014), 20% test (2014–2016) — chronological, no shuffle

02

FinBERT Scoring ★

Primary model: score daily aggregated text — positive, negative, neutral probabilities
+ score = (pos – neg)

04

Lag Features

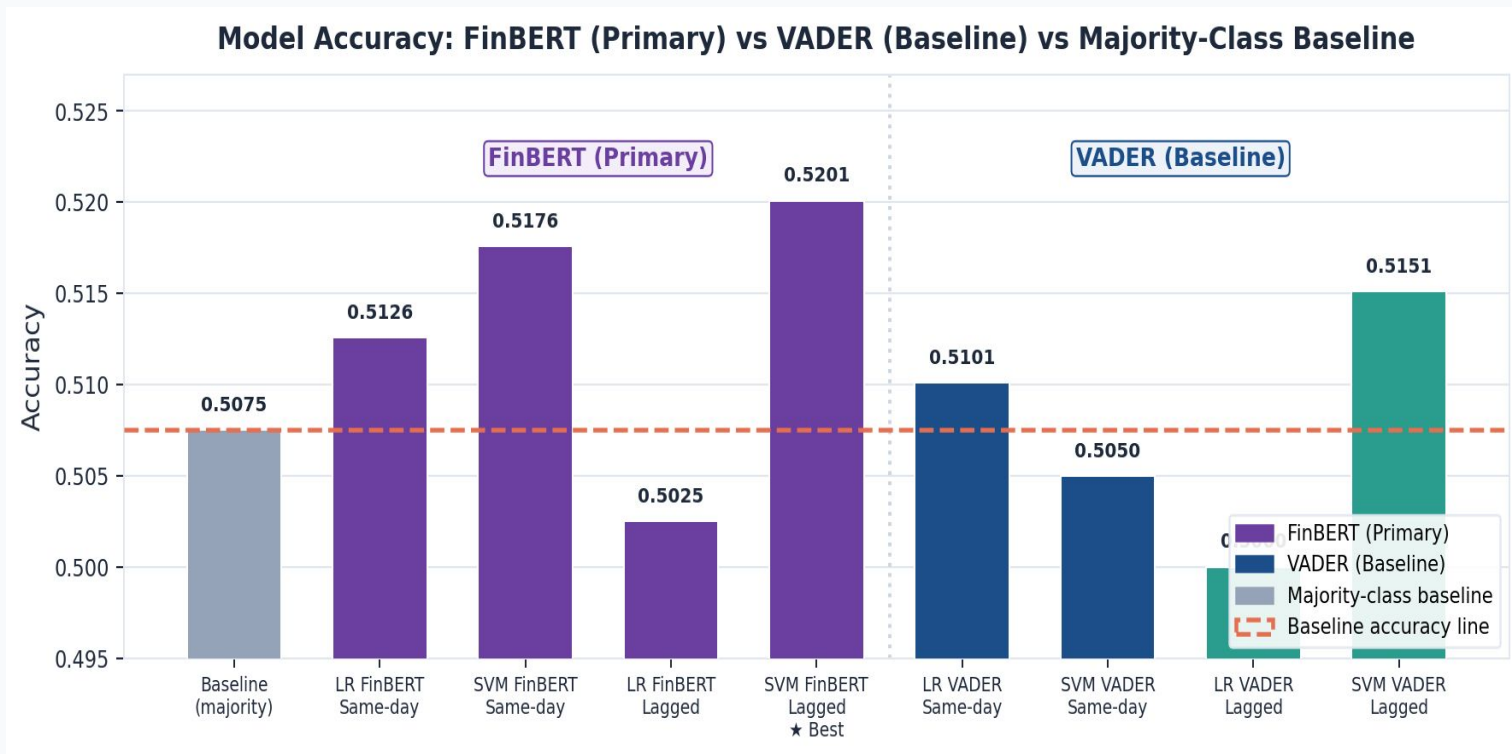
FinBERT and VADER t-1, t-2 lagged scores; rolling 5-day volatility proxy for RQ4

06

Classify & Evaluate

LR + SVM (RBF) on FinBERT (primary) and VADER (baseline) feature sets; Accuracy, Precision, Recall, F1

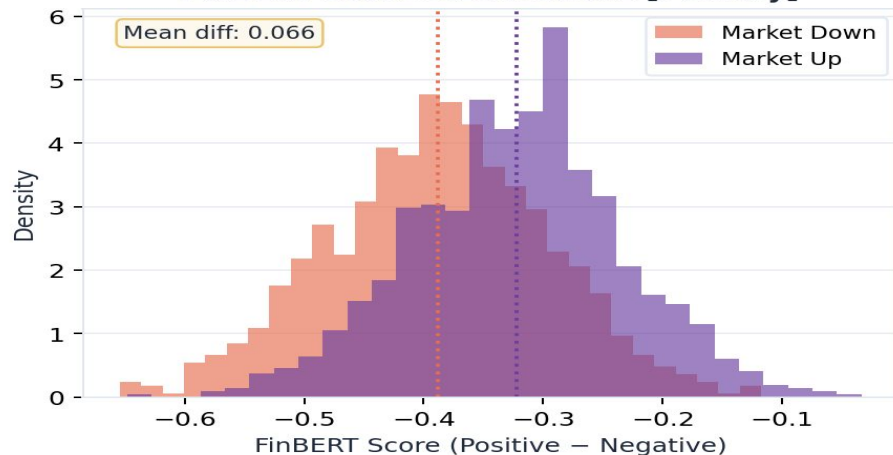
Results



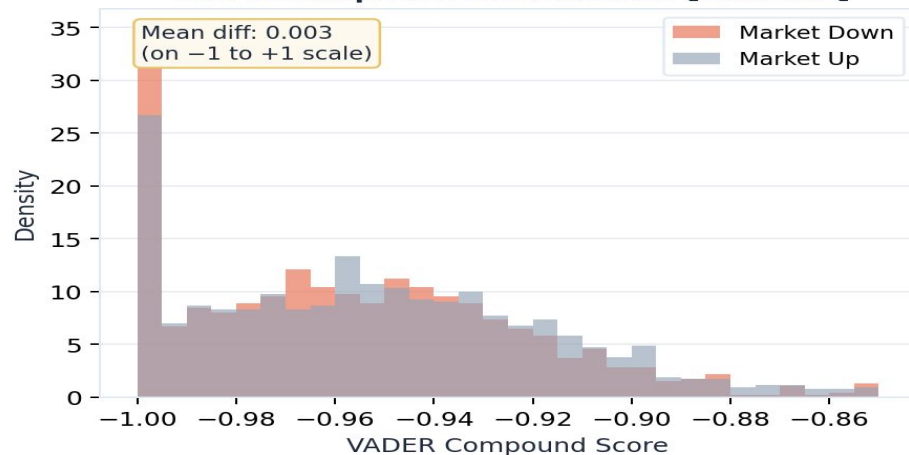
Results

The Data Bottleneck: Both Models See Near-Zero Signal from General News

FinBERT Score Distributions [Primary]



VADER Compound Distributions [Baseline]



52.01%

Best FinBERT model (SVM lagged) —
highest accuracy overall

51.51%

Best VADER model (SVM lagged) —
FinBERT still wins by 0.5 pp

$r \approx 0$

FinBERT score correlation with market
direction — near-zero signal

$p=0.28$

Extreme-day volatility test — not
significant at $\alpha=0.05$

Results: Answering All Five Research Questions

RQ1: Does aggregated sentiment correlate with same-day DJIA direction?

No. FinBERT $r = +0.021$, $p = 0.35$. VADER $r = -0.003$, $p = 0.89$. Both near zero — general news carries no same-day market signal.

RQ2: Can lagged sentiment predict next-day movement above chance?

Marginally. Best: SVM FinBERT lagged = 52.01% vs 50.75% baseline (+1.26 pp). Gain is real but not practically useful.

RQ3: Are negative headlines more influential than positive ones?

Weakly. FinBERT negative probability shows stronger market association than positive. Extreme negative days show 46% up-rate vs 54% for normal days.

RQ4: Do extreme sentiment days precede heightened volatility?

Weak trend, not significant. Extreme FinBERT days: next-day change rate 0.450 vs 0.433 for normal. Mann-Whitney $p = 0.22$ — below $\alpha=0.05$.

RQ5: Does news source domain matter for signal strength?

Yes — this is the central finding. Even FinBERT (the best available financial NLP tool) produces near-zero market signal on *r/worldnews*. The source, not the model, is the bottleneck. Financially-relevant news (Reuters, WSJ) is the necessary next step.

Overarching finding: Even our strongest model (FinBERT) cannot extract market signal from general news. Source selection is as critical as model choice.

Discussion and Key Takeaways

Why Did Even FinBERT Fail to Predict Market Direction?

Wrong Data Source

r/worldnews covers geopolitical crises and disasters — persistently negative regardless of market movement. Neither FinBERT nor VADER can extract signal that does not exist.

Tool–Text Mismatch

FinBERT was trained on financial news — but r/worldnews is not financial news. Even the best financial NLP tool cannot overcome a non-financial source.

Signal Already Priced

If EMH holds, end-of-day sentiment aggregates arrive after prices have adjusted — no exploitable lag remains even with a domain-matched model.

Key Takeaways

- 1 FinBERT outperforms VADER across all configurations — confirming domain-specific encoders add value. Best: SVM FinBERT lagged = 52.01%.
- 2 Negative headlines carry marginally more weight than positive in both models — consistent with loss-aversion theory (RQ3).
- 3 Extreme sentiment days show a weak, non-significant tendency to precede elevated directional volatility (RQ4, $p = 0.22$).